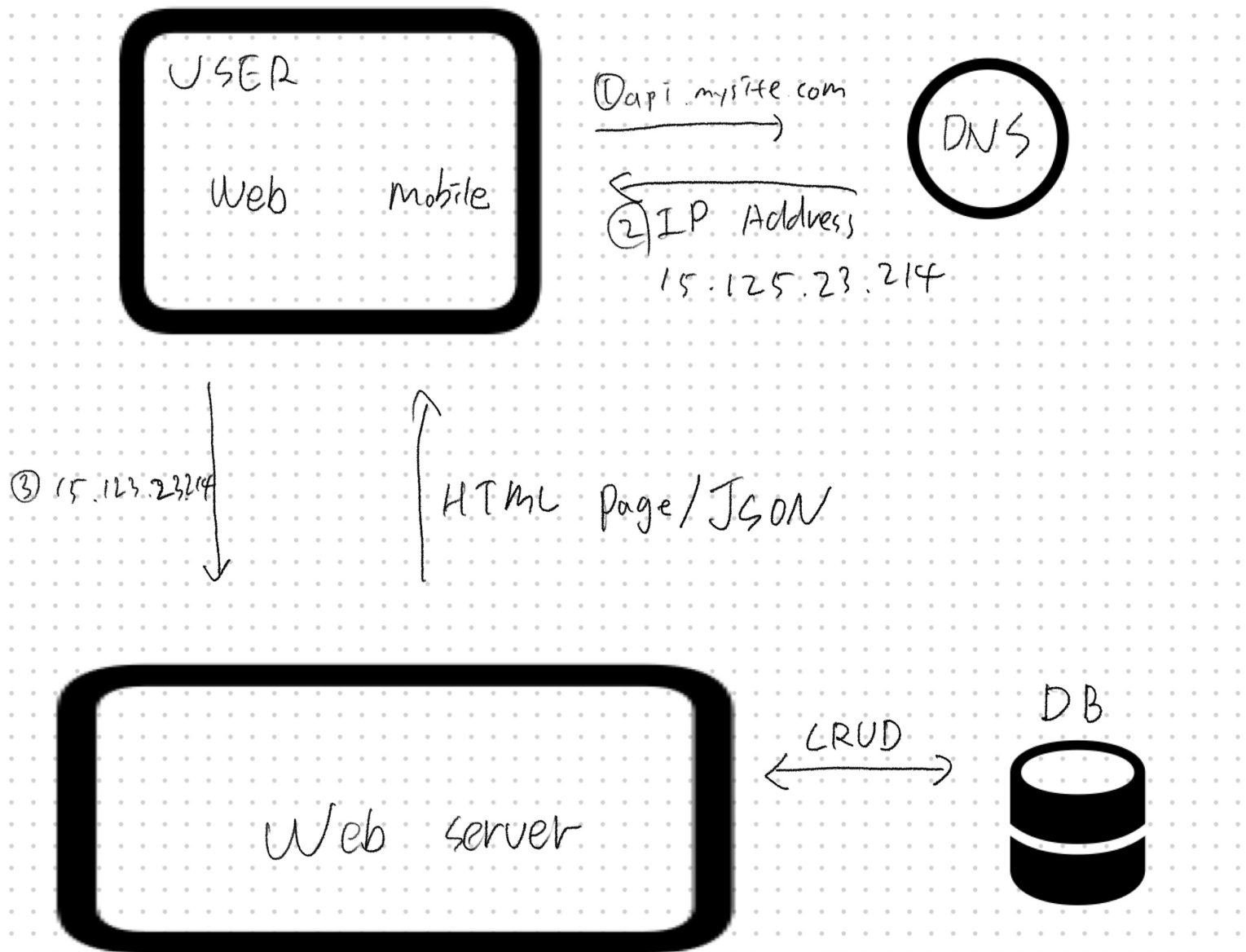


Single Server Setup

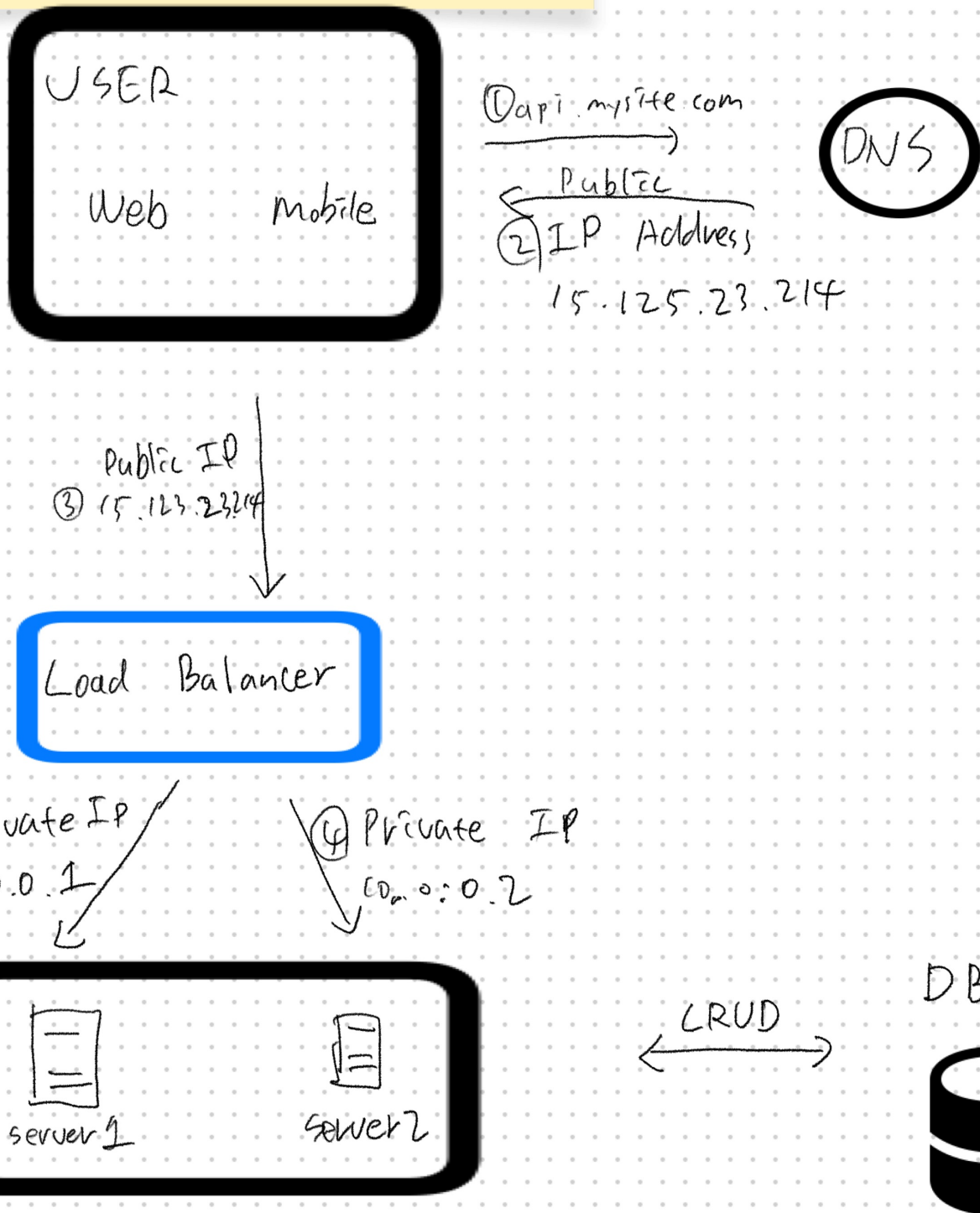


RDBMS vs NoSQL

- Low Latency
- unstructured
- need only rehydrate/de-
- store massive data

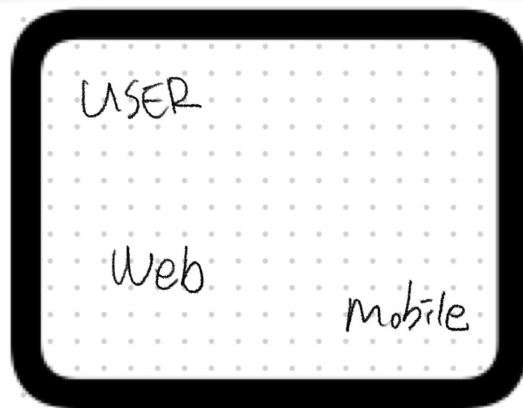
Vertical vs Horizontal

- "scale up"
 - add power (CPU, RAM...)
 - traffic is low
 - simple!
 - Hard limit
 - No failover or redundancy
- "scale-out"
 - add more servers
 - "Large scale"



Database Replication

- Better Performance
- Reliability
- High availability



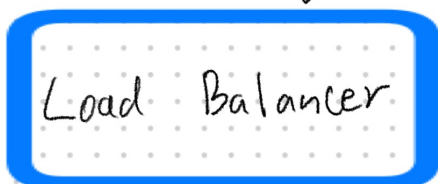
① api.mysite.com

Public
② IP Address

15.125.23.214

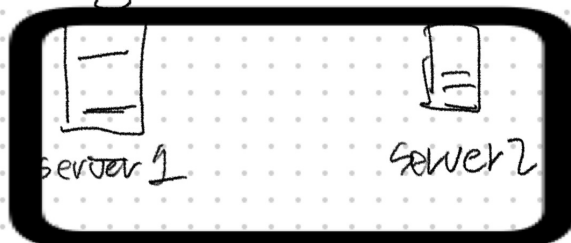


Public IP
③ 15.123.23214

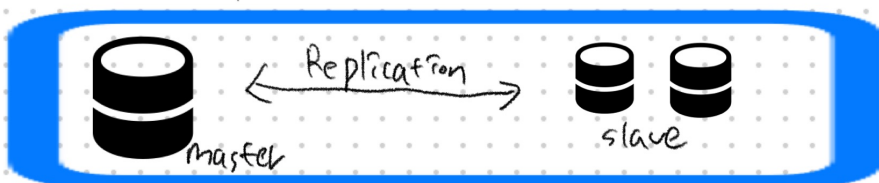
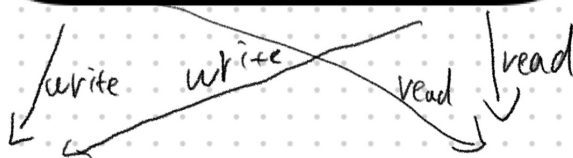


④ Private IP
10.0.0.1

④ Private IP
10.0.0.2



Web tier

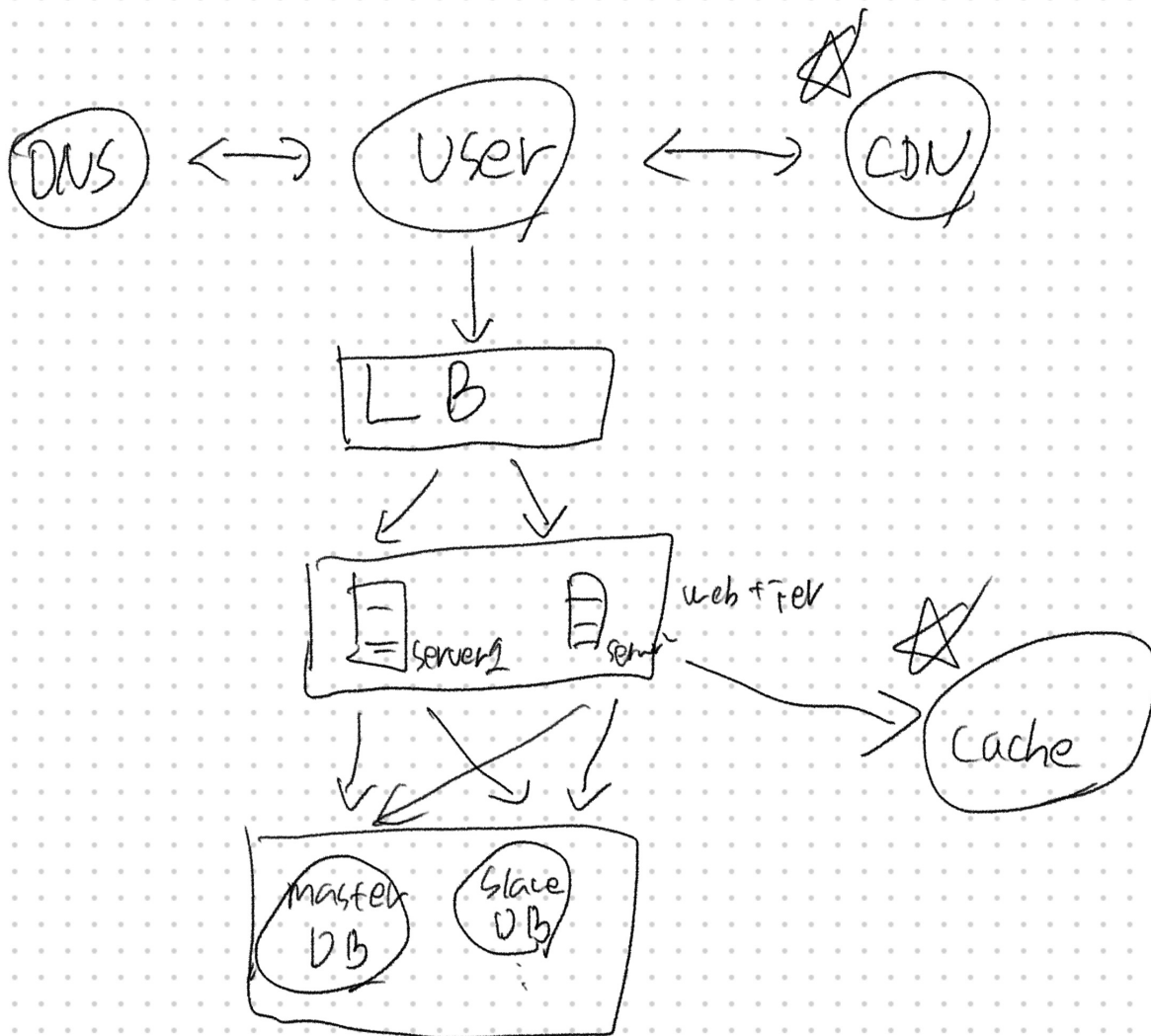


Cache

- Decide when to use
- Expiration policy
- Consistency
- Mitigating failures
- Eviction Policy - LRU, LFU, FIFO

CDN

- Images, videos, CSS, JS ..
- Cost
- Setting cache expiry
- CDN Fallback
- Invalidating files



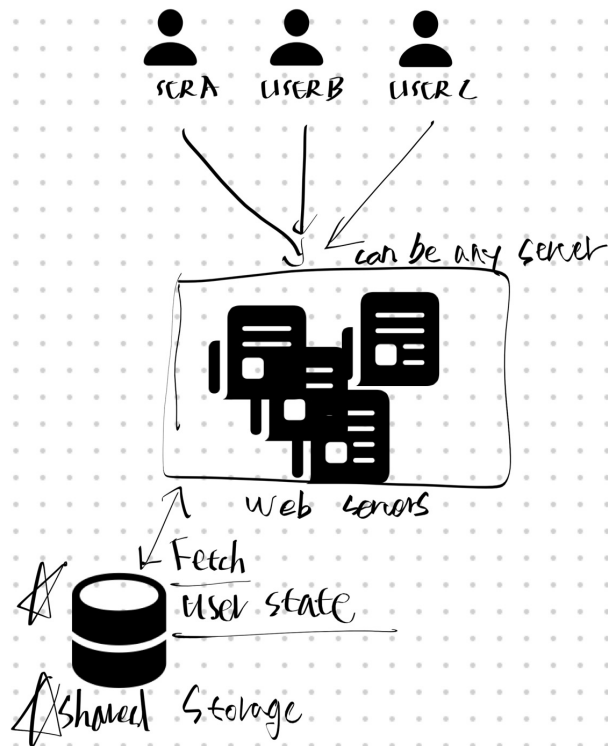
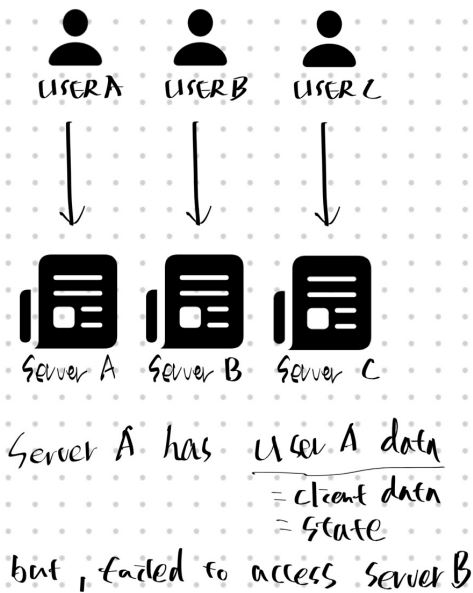
SPOF: A Single Point Of Failure

=> a part of a system that if it fails,
will stop the entire system from working!

Stateful

VS

Stateless



Data Center

